

Sugam Dembla

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WORK EXPERIENCE

National Payments Corporation of India (NPCI) <i>Senior Data Scientist</i>	Hyderabad, Telangana, India <i>Aug 2024 – Present</i>
- Led design and development of LLM-powered Conversational Payments for BHIM & UPI 123Pay across 10 Indian languages, deployed at national scale with ~50k daily hits and <500ms end-to-end latency. - Fine-tuned and deployed Gemma, LLaMA, Mistral, GPT, DeepSeek models for financial tasks (send, balance, transaction history, mobile/FASTag recharge, bill payments etc.) achieving 95%+ intent/entity accuracy across 10 Indian languages. - Developed Language Identification (LID) and Wake-Word Detection (“Hi BHIM”) modules achieving 90%+ accuracy, enabling real-time voice-triggered interactions. - Designed hypothesis-driven experiments, evaluation pipelines (BLEU, ROUGE, accuracy, hallucination checks), and analytics frameworks to measure model impact and prioritize high-value payment flows. - Implemented high-throughput inference pipeline using TensorRT, ONNX Runtime, Triton Server, vLLM, and GGUF quantization, achieving ~40% latency reduction. - Built end-to-end MLOps pipelines (Docker, Kubernetes, FastAPI, Jenkins, Git) for training, evaluation, deployment, and monitoring across secure on-prem data centers.	

Indium Software (Client - Uber) <i>Data Scientist</i>	Bengaluru, Karnataka, India <i>Jul 2022 – Jul 2024</i>
- Designed and deployed 15+ production-grade ML models for Uber’s Intelligent Document Processing (IDP) platform, enabling document verification, classification, and extraction that accelerated user onboarding and significantly improved global compliance accuracy. - Achieved 70% reduction in processing time, 2× reduction in manual review, and 90%+ extraction accuracy, with 35% documents reaching 99.5% accuracy. - Conducted A/B tests, regression modeling, and statistical evaluations to validate model upgrades and quantify improvements in onboarding efficiency. - Automated object detection & classification by developing and deploying a YOLO model for Uber ATG and Aurora, reducing manual bounding box creation and labeling efforts by 90%+ for operators. - Built end-to-end ETL pipelines from AWS S3 to Hive, leveraging IAM policies for cross-bucket transfers, and developed analytical dashboards (SQL, Hive, Presto, PySpark) to track KPIs, providing insights and feature rollout. - Enhanced operator performance by 23% by monitoring detailed QA metrics (Precision, Recall, Geometry, Motion, Linking, Metadata) across scene labeling workflows, driving higher data quality and model reliability.	

Capgemini (Client - Swift) <i>Data Analyst</i>	Pune, Maharashtra, India <i>Sep 2021 - Jul 2022</i>
- Built ML-driven platform components to enhance SWIFT system scalability and performance monitoring, contributing to higher feature stability and improved platform responsiveness. - Managed and optimized large-scale relational datasets on SWIFT systems, improving data integrity and storage efficiency while reducing analytical query latency by up to 35% through schema and performance tuning. - Designed and implemented data modeling and warehousing pipelines that improved cross-team analytical efficiency and accelerated insight-generation workflows by 25%. - Developed end-to-end API and UI automation frameworks (Python, Robot Framework), significantly lowering regression risks and reducing manual QA cycles by 30%. - Streamlined development workflows by standardizing Git/Bitbucket practices and automating CI/CD pipelines with Jenkins, reducing deployment errors by >50% and improving release reliability across Agile teams. - Implemented secure, audit-ready deployment workflows aligned with SWIFT compliance standards, ensuring reproducible builds and minimizing operational risk in production environments.	

Ekklavya Infosys and Virtual Analytics <i>Machine Learning Intern</i>	Pune, Maharashtra, India <i>May 2020 - Sep 2021</i>
- Built a Market Basket Analysis system using offline/online time-series retail data, identifying high-frequency item pairs and improving recommendation relevance by ~25%, driving higher personalized offer conversions. - Created analytical dashboards & visualizations using Python & R to extract insights and communicate patterns to teams and clients. - Published two peer-reviewed research papers in NLP and Computer Vision including a Springer Nature paper on contradiction detection in legal proceedings and an IRJET paper on drone-based object detection for safety & surveillance. - Designed an NLP/NLU contradiction-detection pipeline for the legal domain, benchmarking ELMo, BERT, RoBERTa, XLNet, and ALBERT, achieving 88% accuracy with ALBERT and outperforming baseline Transformer models. - Developed a drone-based real-time object detection and surveillance pipeline by benchmarking YOLO, CNN, Fast-CNN, and Faster-CNN achieving 5× to 10× higher FPS throughput, ~4× lower inference latency, and a measurable improvement in end-to-end detection accuracy through model and pipeline optimization for safety and monitoring applications.	

EDUCATION

Symbiosis Skills and Professional University <i>Bachelor of Technology (B.Tech) - 9.02 CGPA</i>	Pune, Maharashtra, India <i>Graduation Date: May 2021</i>
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SKILLS

Python, SQL, Data Science, Machine Learning, GenAI, LLMs, Natural Language Processing (NLP), Computer Vision (CV), Wake Word Detection, Language Identification (LID), Automatic Speech Recognition (ASR), NMT, Deep Learning, Multilingual Transformers, LLaMA, Mistral, Gemma, GPT, vLLM, LoRA, HuggingFace, Unsloth, RAG, ONNX, TensorRT, Triton, GGUF quantization, Spark, PySpark, Hadoop, Kafka, Hive, Presto, Docker, Kubernetes, FastAPI, Flask, Jenkins, Git, CI/CD, AWS, GCP, On-prem GPU clusters, A/B Testing, Offline Evaluation, Metric Dashboards, Tableau, Looker, Statistical Modeling, Experiment Design, Hypothesis Testing.

PUBLICATIONS

- Identifying Contradictions in the Legal Proceedings Using Natural Language Models** [[Link](#)]
 - *Springer Nature Computer Science*
 - DOI - <https://doi.org/10.1007/s42979-022-01075-3>
- Machine Learning-based Object Detection and Classification using Drone** [[Link](#)]
 - *International Research Journal of Engineering and Technology (IRJET)*
 - Volume: 08 Issue: 06 | e-ISSN: 2395-0056 | p-ISSN: 2395-0072